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Satellite Imagery Analysis for Crop Type Segmentation Using U-Net Architecture

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Abstract

Analyzing satellite images plays a critical role in surveying and monitoring agricultural areas, enabling various applications such as precision agriculture, land use planning, and yield estimation. One of the crucial steps in these applications is accurate crop-type segmentation from satellite imagery. This task becomes challenging in the scenarios of smallholder farms due to their irregular field shapes, frequent cloud cover, small plot sizes, and a severe shortage of training data, which make it difficult to apply conventional machine learning methods effectively. In this research work, a technique for segmenting the crop types from the difficult scenario of smallholder farms based on fully convolutional encoder-decoder semantic segmentation architecture, U-Net, has been proposed and its performance has been compared with the traditional machine learning techniques. To evaluate the proposed approach, experimental assessments were conducted on the Kenya satellite imagery crop dataset. The proposed technique achieved an accuracy, precision, recall, and F1 score of 95.3%, 80.2%, 68.1%, and 73.6%, respectively. The results demonstrate that the U-Net model surpasses conventional image classification methods in accurately segmenting different crop types.

Keywords: convolutional neural networks; crop type mapping; agriculture; satellite images; Sentinel-2; fully convolutional networks; semantic image segmentation

1. Introduction

The food supply system is under increasing pressure because of the possibility of human population expansion; it is predicted that there will be 8.7 billion people on the planet by 2030. Additionally, agricultural output is already being hampered by climate change effects and severe natural catastrophes (such as drought and flood) that pose a danger to global food security [1]. Therefore, it is essential to gather accurate data about the location, scope, kind, health, and yield of crops in order to provide food security, reduce poverty, and manage water resources. In order to ensure that high-quality information about crops is produced to reach certain goals, it is imperative to use effective methodologies, such as sophisticated machine learning and remote sensing (RS) data.

Crop-type maps that are spatially disaggregated are required to measure agricultural diversification and forecast yields. Departments of agriculture in high-income nations generate crop-type maps every year, but they are not yet accessible for middle- and low-income nations [2]. Small fields, few ground truth labels, intercropping, and very varied landscapes are obstacles to mapping crop kinds in smallholder regions.

Generating accurate crop type maps requires the exact classification of every pixel in an image to a crop class. The field of study, known as semantic image segmentation, deals with classifying each pixel in an image into one of the meaningful classes. Traditional techniques for image segmentation employ knowledge-based methods which

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need a lot of human engineering, computational work, and domain expertise. With the introduction of deep learning, these problems with traditional methods are solved using trainable segmentation techniques that model the domain knowledge from a dataset of labeled pixels automatically in an end-to-end manner. Due to their high levels of automation, deep learning-based crop-detection and segmentation methods using high-resolution satellite images have caught the attention of researchers.

In this research work, a methodology to employ remotely sensed satellite imagery time-series to generate crop type maps in the difficult scenario of smallholder farms is proposed. Specifically, the various steps employed to generate crop type maps include removing clouds from satellite scenes, calculating Normalized Difference Vegetation Index (NDVI), temporal interpolation of satellite imagery time-series to fill the missing scenes, converting the problem of crop type classification into crop type segmentation by converting vector crop type masks into raster ground truth maps, and employing a U-Net based encoder-decoder framework for generating crop type segmentation maps. Thus, each pixel is classified into one of several possible crop types given a temporal series of satellite data over an agricultural area. The dataset employed is the openly accessible CV4A Kenya crop type detection. The reason for the selection of this dataset is that automatic crop type categorization presents a special issue in developing nations because smallholder farms, such as those in Africa, typically have smaller fields and fewer ground truth labels than bigger studies carried out in regions like the US. Sparse labels introduce missing data gaps, while smaller fields provide fewer pixels of information. In addition, rain and cloud cover predominate during the growth season, which reduces optical imagery vision. Accurate crop type segmentation for such scenarios may aid in understanding how farmers choose which crops to cultivate, shed light on how different crop types interact with the environment, provide details on the diversity of crops and their effects on human nutrition, and make crop monitoring and yield estimation easier. The findings in terms of accuracy, precision, recall, and F1 score demonstrate that trustworthy and solid conclusions were obtained and that the proposed model may be utilized successfully to correctly segment crops.

The contributions of this research work can be summarized as follows:

- Using a multi-crop Sentinel-2 satellite imagery time-series dataset, an effective crop type segmentation model was proposed for the challenging situation of smallholder farms in developing nations.
- The proposed methodology employed various steps employed to generate crop type maps include removing clouds from satellite scenes, calculating NDVI, temporal interpolation of satellite imagery time-series to fill the missing scenes, converting the problem of crop type classification into crop type segmentation by converting vector crop type masks into raster ground truth maps, and employing a U-Net based encoder-decoder framework for generating crop type segmentation maps.
- The performance of the proposed model was compared with various existing machine learning based classifications models and deep learning-based segmentation models.

The remainder of the paper is organized as follows. A summary of prior research works in this area is given in Section 2. Section 3 explains the proposed methodology. The results and analyses of the experimental dataset are condensed in Section 4. Finally, the proposed work is concluded in Section 5 along with future research pointers.

2. Literature review

This section examines the various remote sensing picture crop segmentation techniques and algorithms across time. Deep learning-based techniques have made great strides in recent years. One of the fastest-growing areas of machine learning, deep learning has been developing useful tools that have produced many accurate results [4]. New and efficient architectures are proposed, and state-of-the-art performance has shown a notable improvement on several segmentation benchmarks [5]. Various techniques have been used to explore crop segmentation of remote sensing images, however even after two decades of research, this field is still in its infancy.

In areas lacking crop labels at the field level, Kluger et al. [6], suggest a way to improve satellite-based crop mapping. By making changes in crop distribution and feature transformations across regions into account, it adjusts the classifier using aggregate-level crop statistics. The method involves accounting for shifts in the mean feature vector and reweighting the classifier's posterior probabilities based on class prevalence differences between training and test sets. Using Linear Discriminant Analysis and other classifiers like Random Forests, the methodology showed considerable improvements in accuracy.

Du et al. present [7] a deep learning approach for precisely mapping crop areas based on high-resolution remote sensing images. In order to categorize and extract crop areas, the method uses a deep semantic segmentation network, resulting in a high overall accuracy of 95% and a Kappa coefficient of 0.90. The suggested solution shows scalability across diverse crop kinds and outperforms previous deep semantic segmentation networks and conventional machine learning algorithms. It handles fine features in RGB high spatial resolution photos and successfully represents the small, asymmetrical fields of smallholder agriculture.

Rustowicz et al. [8] focus on mapping crop types in Ghana and South Sudan using remote sensing and deep learning. The authors use a 3D U-Net and a hybrid Convolutional Neural Network (CNN) – Recurrent Neural Network (RNN) model to achieve respectable performance despite obstacles like few and poor-quality data. Studies on ablation are carried out, and comparisons with a random forest baseline are made. The authors draw attention to the fact that precise crop-type maps can provide insightful analyses of food systems, especially in developing nations with rare ground surveys.

Asam et al. [9], using Sentinel-1, Sentinel-2, and LPIS data, describe a method for mapping crop types in Germany. Using Random Forest, classification was accomplished with a 75.5% total map accuracy. For winter wheat, maize, winter rapeseed, and sugar beet, high accuracy was attained. The method came quite close to the actual crop class areas, differing from the official agricultural census data by barely 1%. The technique has the potential to be used repeatedly to map crop types across the country.

Defourny et al. [10] introduce Sen2-Agri, a system for monitoring crops on a national scale utilizing data from Sentinel-2 and Landsat 8. It produces vegetation indicators, farmland masks, crop-type maps, and composites without clouds. Large amounts of time series data are processed using machine learning techniques and in situ data. In five local sites across three nations, the system was successfully tested. Most nations can monitor agriculture thanks to the open-source Sen2-Agri technology, which resolves issues with widespread high-resolution crop monitoring.

With a focus on sub-Saharan Africa, Kerner et al. [11], underline the necessity of precise crop type mapping for food security. Their study presents a brand-new dataset of ground-truth crop types and field boundaries in western Kenya that was gathered over the course of an unusual growing season. With up to 64% for maize and 70% for cassava, the authors use k Nearest Neighbours to illustrate the accuracy of crop type categorization. To enhance crop class separability, they advocate incorporating more earth observation data. The purpose of their work was to stimulate additional studies on crop type mapping for smallholder agriculture.

Li et al. [12] describe a technique for mapping crop kinds in China's Hetao Irrigation district using deep learning and Sentinel-2 imagery. With the use of a feature selection strategy, the U-Net algorithm can classify large crops more effectively and with higher accuracy. There are certain key times for early crop recognition. Tests of the built models in 2020 and 2021 produced results with a high F1 score, mean intersection-over-union and general correctness. The suggested technique provides a timely and precise tool for crop mapping in the Hetao Irrigation district.

In areas with intricate planting arrangements, Luo et al. [13] emphasize how well Sentinel-2 satellite data and machine learning algorithms work to produce precise crop maps. The study was successful in classifying crops on the North China Plain by applying a variety of predictors and feature sets, including NDVI time series, phenological traits, and textural features. The findings show that phenological and textural characteristics are important in model prediction. Random Forest outperformed all other categorization methods when they were compared. In areas with varied cropping patterns, the study shows the effectiveness of feature selection and machine learning for high-resolution crop mapping.

Prins et al. [14] assessed the application of machine learning algorithms, aerial images, Sentinel-2 imagery, LiDAR data, and Sentinel-2 data for discriminating five crop varieties in an intensively farmed area. According to their study, LiDAR data can successfully distinguish between various crop varieties, with XGBoost having the greatest overall accuracy at 87.8%. The ability to distinguish between crops with significant height differences, such as orchards from groundnuts, was made possible by the LiDAR data. The three datasets worked best together to distinguish between the various types of crops, with RF having the highest overall accuracy (94.4%). These results lay the groundwork for choosing the best set of machine-learning algorithms and remotely sensed data sources for operational crop-type mapping.

Cai et al. [15] used time-series Landsat data and a machine learning methodology to propose a high-performance in-season crop-type categorization system. The study was carried out in Champaign County, Illinois, and the findings demonstrated that a cost-effective viable option for field-level and in-season crop-type classification for a corn/soybean-dominated Corn Belt landscape is provided by combining time-series Landsat Surface Reflectance Data and Common Land Units (CLUs) with a machine learning approach. For each CLU field, the Deep Neural Network (DNN) classification model was used to separate the patches of corn and soybeans. The model was trained and tested using the Crop Data Layer (CDL) as the source of truth. The study discovered that this strategy, which can be scaled up to cover wide geographic areas like the U.S. Corn Belt, has a high overall accuracy of classification (95%). The results point to the potential of the methodology described in this study for the precise, economical, and in-season classification of field-level crop varieties, which may have significant scientific and practical applications.

Seydi et al. [16] suggest a unique deep learning-based method for mapping crop types using Sentinel-2 time-series data. The proposed method performed better than existing cutting-edge classification techniques and obtained high overall accuracy. The study comes to the conclusion that crop mapping must be done promptly and accurately if management and choices are to support food security.

In order to simulate the label-poor environment, Wang et al. [17] depleted the models of labels in different states and years, simulating the label-poor setting. The paper investigates the use of random forests transferred across geographic distance and time, as well as unsupervised methods combined with aggregate crop statistics for crop type mapping in the US Midwest. The US Department of Agriculture's Cropland Data Layer (CDL), which provides crop-type labels with a 30 m spatial resolution, was used to validate their methodology. They discovered that random forests with accuracy routinely over 80% transfer to the target region when trained on regions and years with similar growing degree days (GDD). As GDD discrepancies widen, accuracy falls. Although they identify crops less reliably, unsupervised Gaussian mixture models with class labels taken from county-level agricultural statistics do not need field-level labels for training. In states with low crop diversity, like Illinois, Iowa, Indiana, and Nebraska, GMM achieves over 85% accuracy, but in states with great crop diversity, such as North Dakota, South Dakota, Wisconsin, and Michigan, it occasionally performs no better than random. In regions of the world without many or any ground labels, these techniques give choices for field-resolution crop-type mapping. Therefore, this study comes to the conclusion that harmonic coefficients, whether applied in supervised or unsupervised approaches, can successfully separate crop types. When regional crop compositions and growing conditions (as indicated by GDD) are similar, random forests transfer with great accuracy to nearby geographies.

Du et al. [18] suggest using high-resolution remote sensing pictures to map crop patches using deep learning. The suggested method classifies and extracts crop areas from the images using a deep semantic segmentation network. The study finds that the suggested strategy outperforms other deep semantic segmentation networks and conventional machine learning techniques in effectively obtaining acceptable accuracy (overall accuracy = 95%, kappa = 0.90) of crop area classification in the study area. The suggested method can be easily scaled up to accommodate the various crop varieties in an agricultural area. Overall, the suggested method may develop a precise and efficient model that can handle the high degree of detail in RGB high spatial resolution images and accurately describe the small, erratic fields of smallholder agriculture.

Buttar and Sachan [23] proposed a deep learning-based encoder-decoder framework for the semantic segmentation of crop types from Sentinel-2 satellite images. The proposed architecture employed feature fusion techniques and a lightweight channel attention mechanism to get a precise prediction at the pixel level. The experiments were performed on the dataset of smallholder farms along the drought-stricken stretch of the Orange River in South Africa. Besides the spectral bands present in the dataset, various spectral indices were calculated and ablation experiments were performed to get the most useful combinations of spectral bands and indices. The model achieved the best mean intersection over union score of 62.22% and overall accuracy of 76.22% using all the Sentinel-2 bands and spectral indices including NDVI, Normalized Difference Water Index, and Euclidean norm.

3. The proposed methodology

In this section, first, the public dataset, CV4A Kenya crop type detection dataset used for the training and evaluation is described. Then, the various steps involved in the proposed methodology of segmenting crop types from satellite images are explained. The thorough implementation and hyperparameter details used to train the model are then provided.

3.1 Dataset

After inspecting and choosing 4,688 fields from the original ground reference data, the Radiant Earth Foundation and the Plant Village team assembled the CV4A Kenya crop type detection dataset. A training set (3286 in the train set) and a test set (1402 in the test) were created from the dataset. Four tiles are used to catalogue the dataset. Compared to the original Sentinel-2 tiles, these tiles are smaller as they have been trimmed and chipped to fit the region where labels have been gathered.

For each tile, 13 multi-band observations were made over the course of the growing season. Each observation consists of 12 bands from the Sentinel-2 L2A product and a cloud probability layer. The Sentinel-2 atmospheric correction algorithm (Sen2Cor) created the cloud probability layer, which offers an estimated cloud probability (0–100%) per pixel. A standard grid with a spatial resolution of 10 m is used to map all the bands. The images have a resolution of 607x504, and each image is stored as a .tiff file. A .tiff file is a tagged image file with the TIF or TIFF file extension. High-quality raster graphics are stored in this kind of file. The format allows lossless compression, in which no image data is lost during compression. This enables graphic designers and photographers to store their high-quality photographs in a manageable quantity of storage space without sacrificing quality.

The ground truth mask for each image consists of vector polygons, each representing an area covered by a specific type of crop. Seven different crop types are classified: three are pure (maize, cassava, and common bean) and four are intercropped (maize/common bean, cassava/common bean, maize/soybean).

3.2 The proposed segmentation methodology

In this section, the various steps employed to generate crop type maps are detailed. These include removing clouds from satellite scenes, calculating NDVI, temporal interpolation of satellite imagery time-series to fill the missing scenes, converting the problem of crop type classification into crop type segmentation by converting vector crop type masks into raster ground truth maps, and employing a U-Net based encoder-decoder framework for generating crop type segmentation maps. These tasks were executed with the help of eo-learn Python library.

3.2.1 Cloud removal: In this task, cloudy satellite scenes were filtered using the information given by the cloud probability layer available in the dataset along with Sentinel-2 spectral bands. The cloud probabilities were first converted into a binary cloud probability mask, where each pixel was classified into cloudy or non-cloudy using Otsu thresholding. This mask was then used to remove the scenes with more than 70% cloud coverage. This task aids in retaining the regions where trustworthy data are accessible for further processing. Also, a valid data mask was computed which is simply the negation of cloud mask, indicating pixels that contain valid data. This mask was used in further processing during the temporal interpolation step.

3.2.2 Computing NDVI: Another operation involved using the Red and Near Infra-Red (NIR) spectral bands of Sentinel-2 to compute the NDVI, which offers important insights into the health and density of the vegetation. NDVI is calculated as:

$$NDVI = (NIR - Red) \,/\, (NIR + Red). \tag{1}$$

3.2.3 Temporal interpolation: Using the valid data mask computed in step 3.2.1, temporal interpolation was done on the satellite data to ensure a continuous and smooth time-series to fill in any gaps or missing data at the start or finish of the time series.

3.2.4 Conversion of the problem of crop type classification into crop type segmentation: In order to approach the crop classification problem as a semantic segmentation task, where crop types are indicated by a raster layer, target masks for both the training and test images were produced by converting the vector polygons representing each crop type into a raster. Fig. 1 shows a satellite imagery tile depicting the RGB composite of an original satellite capture, the cloud probability mask, linearly interpolated RGB, linearly interpolated NDVI, and the ground truth mask representing different crops and fields.

A satellite imagery tile depicting (a) the RGB composite of an original satellite capture, (b) the cloud mask, (c) linearly interpolated RGB, (d) linearly interpolated NDVI, (e) the ground truth mask representing different crops, (f) the ground truth mask representing different fields

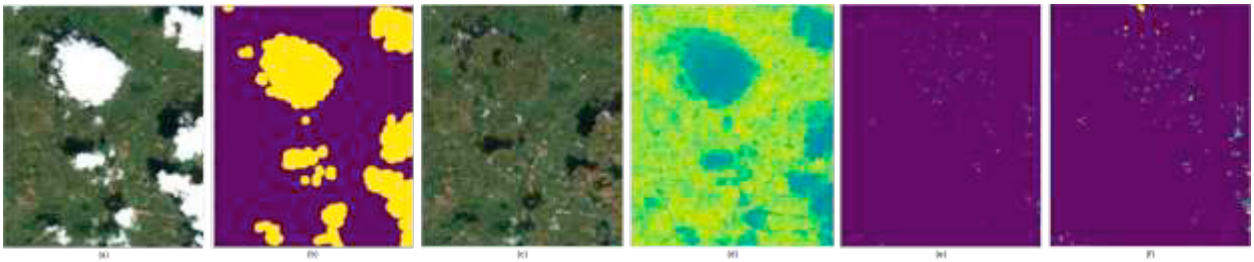

Fig. 1. A satellite imagery tile depicting (a) the RGB composite of an original satellite capture, (b) the cloud mask, (c) linearly interpolated RGB, (d) linearly interpolated NDVI, (e) the ground truth mask representing different crops, (f) the ground truth mask representing different fields.

3.2.5 Training the semantic segmentation architecture: This study employed one of the most popular encoder-decoder-based semantic segmentation architectures: U-Net [19]. U-Net evolved from a traditional CNN; initially proposed by Olaf Ranneberger in 2015 for bio-medical image segmentation, it has been one of the most influential methods to date. This encoder-decoder architecture captures higher-dimensional image representations to make a dense prediction by processing information layer by layer. A label for each pixel is produced when it receives an image as input. U-Net has superior segmentation accuracy and needs a smaller training dataset than other CNNs [3]. The architecture can localize and distinguish borders by classifying every pixel so that the input and the output share the same size. Also, due to the absence of any dense layers and the whole architecture consisting of only convolutional layers, U-Net can accept images of any size.

U-Net follows classical autoencoder architecture and as such it contains two sub-structures, that is, it consists of two symmetrical paths: the contracting path (encoder) and the expansive path (decoder) [20]. The encoder structure follows the traditional stack of convolutional and max-pooling layers to increase the receptive field as it goes through the layers. The decoder structure utilizes transposed convolution layers for up-sampling so that the end dimensions are close to that of the input image. Skip connections are placed between convolution and transposed convolution layers of the same shape to preserve details that would have been lost otherwise.

The encoder is like any standard CNN and is used to capture the context of the image. It extracts a meaningful feature map from an input image, doubles the number of channels at every step, and halves the spatial dimension. By comparison, the decoder is used to up-sample the feature maps, where at every step it doubles the spatial dimension and halves the number of channels. It is used to enable precise localization using transposed 2D convolutions along with regular convolutions, thereby increasing the resolution of the output. Summing both the paths, U-Net consists of a total of 23 convolutional layers. The detailed architecture of U-Net is shown in Fig. 2.

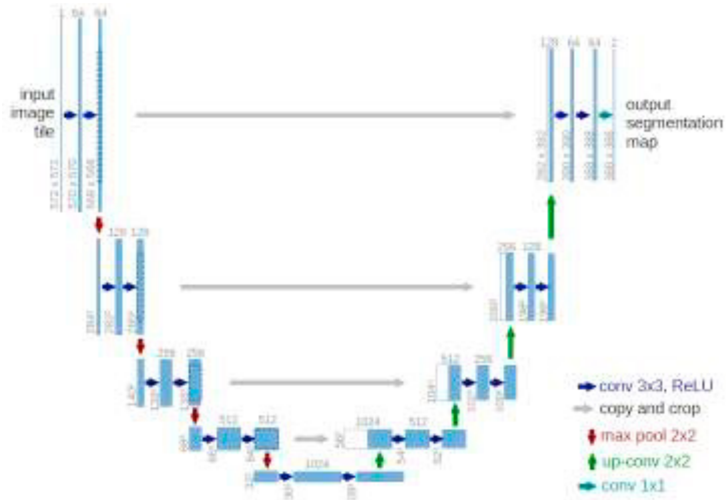

Fig. 2. The architecture of U-Net.

Specifically, the information about image features that the encoder extracts consists of two 3x3 unpadded convolutional layers that are applied repeatedly, followed by rectified linear unit function (ReLU) and 2x2 max-pooling operation with stride 2 which halves the downsizing of the image [21]. This process is done four times, with each down-sampling step doubling the number of feature channels, thereby, allowing the network to learn the "what" information contained in the image. However, in this process, the "where" information is lost; here comes the function of the symmetric decoder part of the architecture. This up-sampling part of the U-Net is an inverted version of the contraction path which employs both transposed and convolutional layers. As a result, the decoder can extract the "where" information.

The advantage of U-Net lies in its addition of symmetric skip connections. The skip connections combine low-level feature maps with high-level ones [3]. This concatenation of the corresponding feature patterns of the encoder and decoder helps improve the accuracy of pixel-level localization, thereby supporting segmentation and providing precise detailed location information. After every concatenation, two regular convolutions are applied, allowing the model to learn how to generate a more accurate output. A final layer, a 1x1 convolution is used to map a 64-component feature vector to reshape the image to satisfy the prediction requirements.

3.3 Implementation details

To implement the model, Python (version: 3.10.12) and the open-source neural network library Pytorch (version: 2.1.0+cu118) were used. The code runs using GPU: T4, with a memory size of 12GB on Google Colaboratory. The flowchart of the entire working process is shown in Fig. 3.

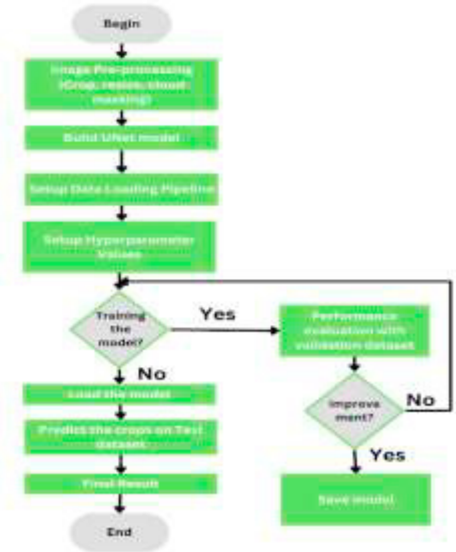

Fig. 3. The entire workflow of the proposed approach

All the satellite imagery tiles were first split into smaller images of size 64x64 pixels for efficient training. The entire dataset was then divided according to 80%-10%-10% split for train, validation, and test.

The Adam optimizer [22] has been used to optimize the model's parameters during the training phase. The learning rate was set to 10^{-3}. Dice loss was chosen as the loss function. The training batch size was set to 16 and the number of epochs was set to 50 as the model began to saturate around the 43^{rd} epoch.

Table 1 provides a summary of all the hyperparameter values used, along with the methods used to construct the suggested model.

Table 1. Proposed Model Implementation Summary.

Specifications	Values
Language	Python
Libraries	Pytorch, FastAI, eo-learn
Device	CUDA
Input image size	64×64×8
Learning rate	10^{-3}
Batch size	16
Optimizer	Adam
Loss function	Dice loss
Activation function	ReLU

4. Experimental results and discussion

To assess the relative efficacy of various segmentation and classification models, quantitative criteria for comparison must be applied. Evaluation metrics, namely accuracy, precision, recall and F1 score, were used for assisting the comparison and evaluation of the proposed model.

Accuracy is a common metric used to measure the overall performance of a classification and segmentation model. It represents the proportion of correctly classified instances (both true positives and true negatives) out of the total number of instances in the dataset. The accuracy formula is:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \tag{2}$$

where, True Positives (TP) is the number of instances that belong to the positive class and are correctly classified as positive by the model. False Positives (FP) is the number of instances that belong to the negative class but are incorrectly classified as positive by the model. True Negatives (TN) is the number of instances that belong to the negative class and are correctly classified as negative by the model. False Negatives (FN) is the number of instances that belong to the positive class but are incorrectly classified as negative by the model.

Precision is a measure of the accuracy of positive predictions made by the model. It answers the question: "Of all the instances predicted as positive, how many were actually positive?" In other words, precision quantifies the proportion of true positive predictions among all positive predictions made by the model. The precision formula is:

$$Precision = TP / (TP + FP) \tag{3}$$

where TP is the number of true positive predictions, and FP is the number of false positive predictions.

Recall (Sensitivity or True Positive Rate) is a measure of how effectively the model can capture all the positive instances present in the dataset. It answers the question: "Of all the actual positive instances, how many were correctly identified by the model?" In other words, recall quantifies the proportion of true positive predictions among all actual positive instances in the dataset. The recall formula is:

$$Recall = TP / (TP + FN) \tag{4}$$

where TP is the number of true positive predictions, and FN is the number of false negative predictions.

The F1 score is a harmonic mean of precision and recall, which helps balance both metrics and provides a single score that represents the overall performance of the model. It is especially useful when there is an uneven class distribution. The F1 score formula is:

$$F1\,Score = 2 * \left(\frac{Precision * Recall}{Precision + Recall}\right) \tag{5}$$

The evaluation of the model's performance on the test dataset is recorded in Table 2. The model achieved an overall accuracy of 95.3% and an F1 score of 73.6%. The outcomes of the proposed approach were also compared with existing machine learning classification models and deep learning-based semantic segmentation models. For comparison with machine learning models, Random Forest, CatBoost, and KNN were used. Fully Convolutional Network (FCN) and SegNet models are the deep learning-based encoder-decoder segmentation models used for performance comparison. From Table 2, it can be seen that the proposed model, when compared to other models, achieved better accuracy, lagging the best competitor, i. e., FCN behind by 4.34% on F1 score metric, thus demonstrating the effectiveness of U-Net. It can also be seen that the deep learning models, U-Net, FCN, and SegNet, demonstrated better performance on crop type segmentation as compared to machine learning models. As a result, it is established that the implemented model presented enhanced results and achieved better accuracy.

Table 2. Performance of the proposed approach on various evaluation metrics.

	Accuracy (%)	Precision (%)	Recall (%)	F1 score
Proposed approach using U-Net architecture	95.3	80.2	68.1	73.6
FCN	88.08	75.54	63.95	69.26
SegNet	81.27	71.45	56.86	63.33
KNN	70.31	52.7	45.6	48.89
CatBoost	68.26	49.5	37.6	42.74
Random forest	50.18	34.4	28.8	31.35

Table 3 presents the performance of the proposed approach on individual crops in the dataset. As seen from Table 3, the proposed model achieves the maximum accuracy up to 70% for common bean, 69% for maize, and 64% for cassava. These three classes represent single crops. On the contrary, the performance of the model is low on other classes representing intercropping, i. e., “maize and common bean”, “maize and cassava”, “maize and soybean”, and “cassava and common bean”.

Table 3. Performance of the proposed approach on individual crops in the dataset.

Crop type class	Accuracy (%)
Maize	69.1
Cassava	64.0
Common bean	70.4
Maize and common bean	24.9
Maize and cassava	22.0
Maize and soybean	19.4
Cassava and common bean	15.6

5. Conclusions and future work

In this research work, a methodology for segmenting crop types from smallholder farms using Sentinel-2 satellite imagery was proposed. The proposed methodology employed various steps employed to generate crop type maps include removing clouds from satellite scenes, calculating NDVI, temporal interpolation of satellite imagery time-series to fill the missing scenes, converting the problem of crop type classification into crop type segmentation by converting vector crop type masks into raster ground truth maps, and employing a U-Net based encoder-decoder framework for generating crop type segmentation maps. The model was trained on the CV4A Kenya crop detection dataset, and the outcomes were compared to those obtained using the currently used machine learning methods as well as deep learning based semantic segmentation models. The proposed technique achieved accuracy, precision, recall, and F1 score of 95.3%, 80.2%, 68.1%, and 73.6%, respectively, lagging the best competitor, i.e., FCN behind by 4.34% on F1 score metric. Deep learning models, U-Net, FCN, and SegNet, demonstrated better performance on crop type segmentation as compared to machine learning models. This study demonstrates that fully convolutional encoder-decoder based segmentation approaches have great promise for the analysis of satellite imagery for crop segmentation in the future. Crop type detection algorithms calibrated for one region cannot be readily extended to another due to the location-specific character of crop phenotypic and phenology information as well as the variability in canopy-level spectral reflectance among varied settings and management practices. As a future research direction, the proposed approach may be tested for transferability on additional study sites with various geographic characteristics. To reduce the error near field boundaries, one may employ a more complex loss function, such as boundary loss. Additionally, a model for predicting yield can be developed using the crop fields that have been observed.

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